Trade duration, volatility and market impact

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Abstract

We perform an empirical investigation of 'market impact' of trades using a large dataset of transactions executed by institutional investors in the US equity market. We find that price variations during trade execution are mainly driven by the *aggregate order flow imbalance* rather than the direction or size of individual trades. We find the main determinants of the amplitude of these price variations to be market volatility and trade duration. In contrast, trade size and execution speed, as measured by the participation rate, are found to have little or no influence on 'market impact' for orderly trade executions.

Conditional on trade duration, trade size is found to have little influence on price variations during execution. We find evidence for a square-root dependence of price changes on *duration* rather than trade size and propose a simple explanation for this dependence in terms of the well-known square-root scaling of volatility as a function of duration. Our explanation is consistent with previous empirical studies on market impact and provides a simple rationale for the ubiquity of the 'square-root law' in these studies.

We also examine the role of the participation rate in determining 'market impact': using evidence from large VWAP trades with high participation rates, we show that conditional on duration, even large changes in participation rate have negligible influence on market impact, contradicting the assumption, common in optimal execution models, that impact increases with the participation rate. In fact, we provide evidence for the opposite effect: for a given trade size, the *slower* the execution, the *higher* the amplitude of price variations during the trade.

Our findings highlight the need to revisit some common models of market impact and their use in the design of optimal execution, and suggest that it is more meaningful to focus on the modelling of aggregate order flow dynamics and the management of portfolio volatility during execution rather than the optimisation of 'impact' at a trade-by-trade level.

Keywords: Market impact; Trade execution; Liquidity; High-frequency data; Market depth; Trade duration; Order flow imbalance; Volatility.

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1. Introduction

1.1. Trade execution and market impact

The analysis of trade execution costs is a key ingredient in the design and performance measurement of trading strategies. A common approach has been to decompose execution costs into two components: *market impact* and *volatility risk* (Collins and Fabozzi, 1991). Market impact refers to the fact that a trader's order might move the price in an unfavourable direction. Volatility risk refers to possibility of adverse moves in market prices during execution, regardless of the trader's action. It is often assumed that a faster execution leads to higher market impact whereas a slower execution leads to a longer trade duration and therefore a higher exposure to market volatility.

Starting with the seminal paper of Almgren and Chriss (2001), the problem of minimising execution costs has been studied through the angle of this trade-off between market impact and volatility risk. With a few notable exceptions (Almgren et al., 2005), the prevailing view in the literature is that the market impact of a trade, rather than price volatility, is the dominant component of execution costs. In particular, the *size* of the trade and the *speed* at which the trade is executed are often presented as the main variables which determine execution costs. The empirical literature on the *square-root law* of market impact (Almgren et al., 2005; Moro et al., 2009; Toth et al., 2011; Bacry et al., 2015; Donier and Bonart, 2015; Gomes and Waelbroeck, 2015; Zarinelli et al., 2015) points to the *size* of the trade as the key variable determining market impact, while the (mostly theoretical) literature on optimal execution, pioneered by Almgren and Chriss (2001), considers the *speed* at which the trade is executed, and the related concept of *participation rate*, as the key determinants of market impact.

Despite the wide use of such models, the empirical evidence underlying their justification is not entirely convincing for a number of reasons. First, most of the empirical work relating price changes with trade size (Torre, 1999; Moro et al., 2009; Gomes and Waelbroeck, 2015; Bacry et al., 2015; Toth et al., 2016) or participation rate (Almgren and Chriss, 2001; Bacry et al., 2015) is based on the analysis of *unconditional* relations. Notably, the importance of *trade duration* as a key conditioning variable has been mostly left out. Second, with a few exceptions (Easley et al., 2015; Bechler and Ludkovski, 2015), the execution horizon is treated as *exogenous* to the trade size, whereas data on institutional trade executions suggests that they are strongly correlated. Third, although there is strong empirical evidence for the influence of aggregate net order flow on market impact of trades (Hasbrouck, 1991, 2007; Cont et al., 2014), the aforementioned empirical studies do not condition on the net order flow.

Finally, with the exception of Bacry et al. (2015) and Almgren et al. (2005), these empirical studies measure 'market 'impact' by averaging over many trades with similar size, in an attempt to average out the contribution of volatility. This notion of average impact may be relevant for a market participants executing a large number of trades sequentially across 'independent' market scenarios. But for an institutional investor who executes infrequent but large trades, the observed impact may deviate significantly from such an average.

1.2. Contributions

In the present study, we provide a different perspective on the concept of market impact and the determinants of execution costs, emphasising in particular the key role played by the trade execution horizon or *trade duration*.

Based on a detailed analysis of a large dataset of institutional trade executions in the US equity market, we provide evidence that the dominant component of execution costs for orderly trade executions is not the 'market impact' of trades but price volatility. We show that price changes accompanying trade executions are mainly driven by the *aggregate (net) order flow imbalance* and not by characteristics of individual trades, even when the trade size is large. In particular, we show that a buy (sell) trade may be accompanied by a negative (positive) price change with high probability ($\simeq 40\%$).

A second finding is that the amplitude of price changes during execution is proportional to the square-root of trade *duration*, rather than trade size, a relation which has a simple explanation in term of the scaling of volatility as a function of the time horizon (Bachelier, 1900). Moreover, we observe that, once we condition on trade duration, the dependence of price changes on trade size vanishes almost entirely. We also find that, conditional on trade size, the amplitude of price changes during execution increases with duration, invalidating previous claims in the literature regarding the 'invariance' of impact with respect to trade duration (Bouchaud et al., 2018).

Our study also provides evidence that, conditional on trade duration, the *participation rate* has little or no influence on the amplitude of price changes during execution. In fact, we find that *slower* execution leads to higher execution costs, which is explainable by higher exposure to volatility risk during execution. In particular, we find that for day-long VWAP trades, a large increase in the traded quantity (and in the participation rate) from $\simeq 18.5\%$ to $\simeq 50\%$ of daily volume leaves execution cost almost unchanged, contradicting the idea that the participation rate is the main determinant of execution costs.

Although these results fly in the face of some assumptions commonly used in the literature on optimal trade execution, they are consistent with previous studies of Engle et al. (2012); Bershova and Rakhlin (2013) on trade executions in equity markets and (Salem et al., 2018) on trade execution in the US Treasury market, which similarly underline volatility risk as the main determinant of execution costs.

Our findings have implications for optimal trade execution and transaction costs analysis. They question in particular the received notion of 'price impact of a trade' and challenge the view that 'market impact' should be treated as the main determinant of execution costs. Our findings also question the use of participation rate as the key determinant of impact, a feature common to many optimal execution models. Finally, they suggest that instead of focusing on optimising market 'impact' at a trade-by-trade level, investors should focus on modelling the aggregate dynamics of net order flow and on managing the main source of risk: price volatility.

1.3. Relation with previous literature

The concept of market impact of a trade refers to the notion that a market participant's order may move the price in an unfavourable direction, thereby resulting in a higher execution cost. An early reference is Kyle (1985) who introduced the concept that trades by a market participant may have an impact on the market price. In Kyle's model, this impact was modeled as a linear function of the trade size. Huberman and Stanzl (2004) reinforced this assumption of linearity by showing that, provided liquidity is constant, permanent impact must be linear to prevent arbitrage.

Following these insights, a considerable body of empirical research has investigated the relation between trade size and price variations during trade execution (Chan and Lakonishok, 1993, 1995; Torre, 1997, 1999; Lillo et al., 2003; Lillo and Farmer, 2004; Bouchaud et al., 2009; Toth et al., 2011; Bershova and Rakhlin, 2013; Cont et al., 2014; Bacry et al., 2015; Gomes and Waelbroeck, 2015; Toth et al., 2016). In particular, several studies point to a concave dependence of the amplitude of price variations on trade size, sometimes represented as a square root function. These findings are sometimes referred to as the 'square-root law of market impact' and may be summarised by the following relation between the size Q of a trade and its impact $\Delta p(Q)$ on the asset price (Toth et al., 2011):

$$\Delta p(Q) = \varepsilon Y \sigma_D \sqrt{\frac{Q}{V_D}}, \tag{1}$$

where Δp is the difference between the arrival price at the start of the trade and the (volume-weighted) average execution price ('trade VWAP'), $\varepsilon = 1$ for buy trades and $\varepsilon = -1$ for sells, Y is a constant, σ_D is the daily asset volatility, Q is the trade size (number of shares), and V_D is the daily market trading volume. Pohl et al. (2017) give arguments based on dimensional analysis as to why one should expect a square root dependence.

In fact, there is considerable variation in empirical findings: many studies have pointed to deviations from this square root dependence (Lillo et al., 2003) and propose variations of this model where the square root is replaced by a power law with some exponent $\alpha < 1$. Some of these findings are summarised in Table 1.

However, a closer inspection leads to many questions regarding the methodology and the conclusions of these studies. First, most of the empirical work relating price changes and trade size (Torre, 1997, 1999; Moro et al., 2009; Gomes and Waelbroeck, 2015; Bacry et al., 2015; Toth et al., 2011, 2016) has been based on the analysis of *unconditional* relations, typically using a logarithmic regression, i.e. without conditioning on

Authors	**Exponent** α	**Asset**	**Region**
Lillo *et al.* (2003)	0.10-0.50	Equities	US
Almgren *et al.* (2005)	0.60	Equities	US
Moro *et al.* (2009)	0.64-0.72	Equities	UK
	0.44-0.48	Equities	Spain
Toth *et al.* (2011)	0.50-0.60	Futures	NA
Bacry *et al.* (2015)	0.40-0.45	Equities	Europe
Bershova and Rakhlin (2015)	0.47	Equities	US
Donier and Bonart (2015)	0.50	Bitcoin	BTC/USD
Gomes and Walbroeck (2015)	0.50	Equities	NA
Zarinelli *et al.* (2015)	0.47	Equities	US
Toth *et al.* (2016)	0.50	Options	NA

Table 1: Summary of empirical studies on the (power-law) dependence of 'market impact' on trade size. An exponent of 0.5 corresponds to a square-root dependence.

relevant variables such as the execution horizon or *duration* of the trade, or the net order flow (an exception is Zarinelli et al. (2015) which we discuss below). Bacry et al. (2015) present a joint analysis of the trade size and duration, but do not go as far as examining conditional relations. An exception is Zarinelli et al. (2015), who study market impact of large institutional trades using the same database as the present work (ANcerno); these authors study the joint dependence of price changes on the participation rate and market trading volume and investigate the effect of conditioning on market capitalisation, participation rate and market trading volume (but not execution horizon/ trade duration). A key aspect of our analysis is to focus on such *conditional* relations, in particular conditioning on trade duration, to elicit the role of different variables.

Most of the aforementioned studies attempt to explain price changes by using trade features (such as size) as (sole) explanatory variables. This contrasts with the more traditional view which relates market price fluctuations to *aggregate* fluctuations in supply and demand, represented by the net order flow (see e.g. Blume et al. (1989); Hasbrouck (1991); Brown et al. (1997); Hasbrouck and Seppi (2001); Chordia et al. (2002); Cont et al. (2014)). Any study of the influence of additional variables on market prices needs to account for market variables, in particular the *order flow imbalance* or net order flow as a baseline explanatory variable.

Another feature of these empirical studies, with the exception of Bacry et al. (2015) and Almgren et al. (2005), is to focus on *average* price changes during trade execution, where averaging is done across trades with similar sizes. This notion of 'average impact' may be relevant for a market participant executing a large number of trades sequentially across 'independent' market scenarios. However, this is not always the setting in which the averaging is done in these studies: in many cases, the averaging is done across trades of *different* market participants (Zarinelli et al., 2015), executed across overlapping intervals, making the result difficult to interpret. In particular, for an institutional investor who executes infrequent but large trades, the observed impact may deviate significantly from such an 'average' impact measure. By contrast, in the present study we examine the influence of these variables on price variations on a trade-by-trade basis, not just on average.

The square-root law formula (1) focuses on the influence of trade size but does not consider the rate of trading or how the trade is executed. In fact, if taken at face value, the square root law implies that the impact of a trade is *invariant* with respect to the execution horizon, or duration of the execution i.e. that speeding up the execution by a factor 2 has no influence on its market impact (Bouchaud et al., 2018). This is in contrast with the literature on *optimal trade execution* which, beginning with (Bertsimas and Lo, 1998;

Almgren and Chriss, 2001), considers the *speed of execution* as the key variable. Following the pioneering work of Almgren and Chriss (2001), the speed of execution is often parametrised through the *participation rate*, defined as volume executed per unit time as a fraction of total market volume during the same period.

A common feature of optimal execution models, starting with (Almgren and Chriss, 2001), is the decomposition of execution costs into a 'market impact' term, which increases with the speed of execution or participation rate, and a 'volatility term' which arises from price variations during execution, independently of the execution strategy. It is generally assumed that slower execution reduces the impact component but increases exposure to volatility risk: the optimal execution problem is then formulated as a tradeoff between volatility risk and market impact (Almgren and Chriss, 2001; Alfonsi et al., 2010; Gatheral, 2010; Predoiu et al., 2011; Curato et al., 2017).

Our results point to trade duration, or execution horizon, as the main determinant of the amplitude of price variations during trade execution. The role of duration as a determinant of price variations during trade execution has been also noted in (Dufour and Engle, 2000; Engle et al., 2012; Bershova and Rakhlin, 2013; Easley et al., 2015) and (Salem et al., 2018). Engle et al. (2012) emphasised the temporal dimension of execution costs and the contribution of volatility. Bershova and Rakhlin (2013) analysed US equity trades executed by Alliance Bernstein from January 2009 to June 2011 and find that impact is a square-root duration and report an R^2 of 29% (Bershova and Rakhlin, 2013, Table 5). Salem et al. (2018) report order flow imbalance to be the most important feature for predicting price impact in the US Treasury market and find that *slower* the execution leads to *higher* price impact (Salem et al., 2018, p. 9). These findings are consistent with ours and provide independent corroboration of our conclusions in other markets and data sets. The role of trade duration as a key determinant of execution cost is also emphasised by (Easley et al., 2015; Bechler and Ludkovski, 2015).

1.4. Outline

The remainder of the paper is organised as follows. Section 2 describes our data and define the quantities of interest. Section 3 examines the determinants of the direction of price variations during trade execution, Section 4 examines the amplitude of price variations and their relation to trade size and duration. Section 5 discusses the role of trading speed and participation rate. Section 5 concludes the paper.

2. Data sources and quantities of interest

2.1. The ANcerno database

We use institutional trading data provided by ANcerno Ltd., a consulting firm that monitors trade execution costs for institutional clients.[1] The dataset contains institutional trading data on US equities, non-US equities and fixed-income by pension funds, such as CalPERS, the Commonwealth of Virginia, and the YMCA retirement fund; and money managers, such as MFS, Putnam Investments, Lazard Asset Management and Fidelity. Our empirical analysis focuses on US equity trades between 01-Jan-2008 and 31-Dec-2012.

Hu et al. (2018) and Anand et al. (2012) provide a detailed description of the ANcerno dataset. According to Hu et al. (2018), ANcerno Ltd. provides consultancy and advisory services to 1,000+ institutional clients representing over 12% of CRSP volume, higher than the previous estimate of $8-10\%$ provided by Puckett

[1]The ANcerno database has been used in several academic papers, such as Anand et al. (2012, 2013); Chakrabarty et al. (2017); Chemmanur et al. (2009); Chicayachantana et al. (2017); Hu (2009); Zarinelli et al. (2015). For a complete list of papers, description and history of the dataset see Hu et al. (2018).

and Yan (2011). Concerning the quality of the data, Anand et al. (2012) argues that survivorship bias is not an issue for two reasons. First, ANcerno Ltd. representatives have stated that there is no such bias. Second, not all institutions are present at the end of their sample period suggesting that the ANcerno data does not contain only surviving institutions (Anand et al., 2012). Selection bias is also investigated, where ANcerno Ltd. clients might systematically differ from the typical institution, and rejected as there are no explicit client requirements (e.g. dollar size of fund managed, type of institution or number of trades executed) and stocks traded by ANcerno institutions do not statistically differ from stocks traded by the average 13F institution. The only notable difference is that ANcerno institutions trade in larger sizes (Anand et al., 2012).

The ANcerno database contains an order identifier variable which allows us reconstruct the sequence of transactions involved in the execution of a trade.[2] Here we refer to the collection of transactions executed with the same order identifier as a *trade*[3] and to each individual transaction that is part of the same trade as an *order* (sometimes referred to as a 'child order' in the literature). We focus on a subset of trades with the following criteria:

1. We focus on stocks in the Russell 3000 Index, for which intraday data is available for estimating volatility and order book depth.
2. We retain trades with durations lasting at least two minutes and composed of at least five orders.
3. We do not consider overnight trades: only trades executed with start and end time within the same trading day are considered;
4. We only consider transactions execution time before 4:01 PM to avoid estimation biases associated with closing trades.

For each trade we observe, among other variables, the ticker of the stock traded, the trade direction ($\varepsilon = \pm 1$), the placement time (t_s), the execution time of the last order (t_e), the placement price (p_s), the volume-weighted average execution price (p_{vwap}), the volume of the trade (Q) and the volume of each order i (v_i) that constitutes such trade, the daily volume of the traded stock from open to close (V_D), the volume of the traded stock during execution (V_P), the daily highest (p_h) and lowest (p_l) prices. These quantities are reported at one-minute frequency. We define the following variables:

$$
\begin{aligned}
\textit{Trade Duration} \quad (T) \quad &\equiv \quad t_e - t_s, &(2)\\
\textit{Price Change} \quad (\Delta p) \quad &\equiv \quad \frac{(p_{vwap} - p_s)}{p_s}, &(3)\\
\textit{Daily Volume Fraction} \quad &\equiv \quad \frac{Q}{V_D}, &(4)\\
\textit{Average Participation Rate} \quad \pi \quad &\equiv \quad \frac{Q}{V_P}, &(5)\\
\textit{Daily Volatility} \quad &\equiv \quad \sigma_D &(6)
\end{aligned}
$$

Different estimators may be used for daily volatility σ_D:

- The normalised daily range:

$$\sigma_D = \frac{p_h - p_l}{\frac{1}{2}(p_h + p_l)};$$

[2]Earlier versions of the database also contained a client identifier; see for example Puckett and Yan (2011), Anand et al. (2012) and Hu et al. (2018).

[3]Referred to as 'parents orders' or "meta-orders" in some papers Bouchaud et al. (2009).

- A GARCH(1,1) estimator (Bollerslev, 1986) based on daily returns;
- Realised volatility estimators based on 5-minute intraday returns (Andersen et al., 2003), to account for intraday seasonalities.

2.2. *Other data sources*

To match transaction data with data on market prices and order flow, we supplement the ANcerno data with intraday order book data from LOBSTER, [4] which provides detailed tick by tick limit order book data for NASDAQ stocks. LOBSTER contains all limit order submissions, cancellations and executions on each trading day and all events are time-stamped with millisecond precision (Bibinger et al., 2017), reconstructed from NASDAQ's historical TotalView-ITCH data (Huang and Polak, 2011). The LOBSTER *message* files contain the order type, order ID, size and the price direction (buy/sell). The LOBSTER *orderbook* files contain the bid and ask prices and quantities. In our study, we use "Level I" data from LOBSTER; these are all order-book events at the best bid and ask prices (Huang and Polak, 2011).

From the LOBSTER database, we compute the *order flow imbalance* (OFI) for the time interval $[t_{k-1}, t_k]$ as the volume at the best bid V_k^b, minus the volume at the best ask V_k^a divided by the total volume at the best prices $(V_k^b + V_k^a)$:

$$OFI_k \equiv \frac{V_k^b - V_k^a}{V_k^b + V_k^a}, \tag{7}$$

where:

$$V_k^b = L_k^b - C_k^b - M_k^s, \qquad V_k^a = L_k^s - C_k^s - M_k^b. \tag{8}$$

L_k^b and C_k^b denote the total number of shares of limit buy orders arrived to and cancelled from the current best bid during that time interval, respectively. M_k^b denotes the total size of the marketable buy orders that arrived to the current best ask (Cont et al., 2014). Similarly, for sell orders we have the analogous quantities L_k^s, C_k^s, M_k^s.

The LOBSTER data can also be used to intraday indicators of volatility which take into account the intraday seasonality of returns (Andersen et al., 2003).

2.3. *Descriptive statistics*

After applying the filters mentioned in section 2.1, we end up with 410,792 trades over 1,764 different stocks, corresponding to 10,026,055 trade executions. Figure 1 shows the histograms of trade duration in minutes (left panel), (log) daily volume fraction in percentage (central panel), and (log) participation rate in percentage (right panel) for all trades in our dataset. These trades have 389 different duration values, with an average trade duration of 118.94 minutes and standard deviation of 126.69. We can see from the left panel of Figure 1 that most trades have either short durations or durations corresponding to one trading day. There are 16,710 trades (4.07%) with a duration of 2 minutes, 13,236 trades (3.22%) with a duration of 3 minutes, 10,620 trades (2.59%) with a duration of 4 minutes, 9,519 trades (2.32%) with a duration of 5 minutes, and 10,018 trades (2.44%) with a duration corresponding to a full trading day (389-390 minutes).

From the central panel of Figure 1 we can see that although many trades are small in size, there is a considerable number of trades that exceed 5% of daily volume. In particular, the average daily volume

[4] https://lobsterdata.com.

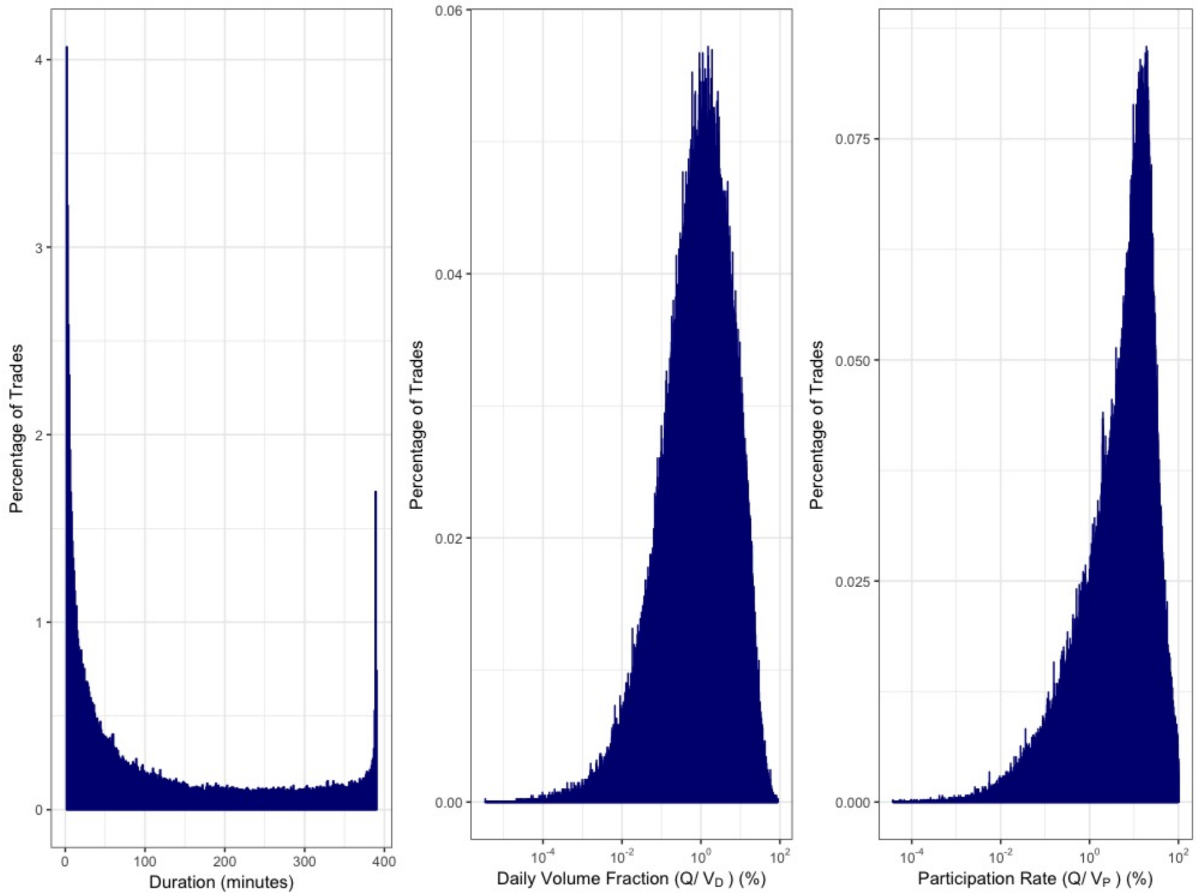

Fig. 1. Histograms of trade duration in minutes (left), (log) daily volume fraction (Q/V_D) in percentage (centre), and (log) participation rate (Q/V_P) in percentage (right), for all trades in ANcerno (2008-2012).

fraction is 3.27%, with a standard deviation of 5.78%, a minimum of 3.5^{-6}% and a maximum of 88.82%. The number of trades with a daily volume fraction equal to or greater than 5% is 78,036 (19.00%); those with a daily volume fraction equal to or greater than 20% is 10,634 (2.59%), and those with a daily volume fraction equal to or greater than 50% is 364 (0.09%).

The right panel of Figure 1, shows a similar story for the participation rate. The average participation rate is 12.39%, with a standard deviation of 14.70%, a minimum of 3.8^{-5}% and a maximum of 98.90%. The number of trades with a participation rate equal to or greater than 5% is 242,855 (59.12%); those with a participation rate equal to or greater than 20% is 85,506 (20.81%), and those with a participation rate equal to or greater than 50% is 12,998 (3.16%). We will devote special attention to large trades, whose participation rates fall in the 4th quartile range, with participation rates of 17.6% or more.

3. Direction of price variations during trade execution

3.1. Do buy trades increase the market price?

One of the assumptions in market impact modelling is that, on average, the price change between the arrival of a trade and its last fill is in the same direction as the trade: buy trades are assumed to push the price up and sell trades to push the price down. For instance, Gomes and Waelbroeck (2015) estimate the average impact under this assumption and define their market impact measure as "the mid-point return from the start to the end of the trading *signed by the metaorder*" (Gomes and Waelbroeck, 2015, notes of Fig. 5, p.784). Similarly, Moro et al. (2009) find that the *average* market impact of trade approximately follows a square-root function of the trade size and is positive for buy trades and negative for sell trades (Moro et al., 2009, eqn. 16, p. 6), while Zarinelli et al. (2015) compute the *average* impact of trades in the ANcerno dataset, after diving the data into equally-populated bins of standardised trade volume (Zarinelli et al., 2015, p. 17).

That a buy (sell) trade should push the price up (down) *on average* seems intuitive, but what many studies fail to report in detail is the level of noise around such average estimates. Moro et al. (2009) point out that "there is a considerable amount of noise in the price change associated with any particular [trade]" (Moro et al., 2009) due to market volatility, while Torre (1999) states that "the cost experience for a single trade will depart, often significantly, from the mean experience". However, the aforementioned studies attempt to 'average out' this effect by averaging price variations over a large number of trades with sizes in a given range (see e.g. Zarinelli et al. (2015)). Almgren et al. (2005), Bershova and Rakhlin (2013) and Bacry et al. (2015) are among the few studies which attempt to quantify the contribution to execution costs of this 'noise' term –which is none other than market volatility– and avoiding taking averages.

If we think of the price change contemporaneous to a trade as being composed of two terms, a *systematic* term due to one's own execution and a random term due to market volatility, then averaging across trades makes sense when analysing a large number of consecutive executions over non-overlapping or 'independent' market scenarios. In this situation it may be argued that the volatility term may average to zero, leaving the systematic 'impact' as the main contribution to the 'average impact'. But this scenario does not hold for institutional market participants in the ANcerno database: most of these market participants execute infrequent but large trades –only a few trades a day– and this notion of 'average impact' is not relevant or meaningful for such market participants. Indeed, as observed in Almgren et al. (2005), in absence of such averaging, volatility is the main contribution to execution costs for such market participants.

We therefore start by studying the *sign* of all trades in our data and by computing the conditional probabilities that price changes are in the same direction of trades. Then, we look at the standard square-root law model of market impact, and test its explanatory power on the observed data without averaging our data. Finally, we analyse and quantify the noise term surrounding the square-root model estimates and draw conclusions about the model performance and the notion of 'price impact of a trade'.

Figure 2 presents a scatter-plot of price changes (standardised by their daily volatility) against the daily volume fraction multiplied by the sign of the trade - positive for buy trades and negative for sell trades. If the assumption that prices moved in the same direction of trades were true, we would only see points in the quadrants where $\Delta p > 0, Q > 0$ and $\Delta p < 0, Q < 0$ and no points in the quadrants defined by $\Delta p > 0, Q < 0$ and $\Delta p < 0, Q > 0$. The scatter-plot clearly shows that this is not the case. Points are distributed across all

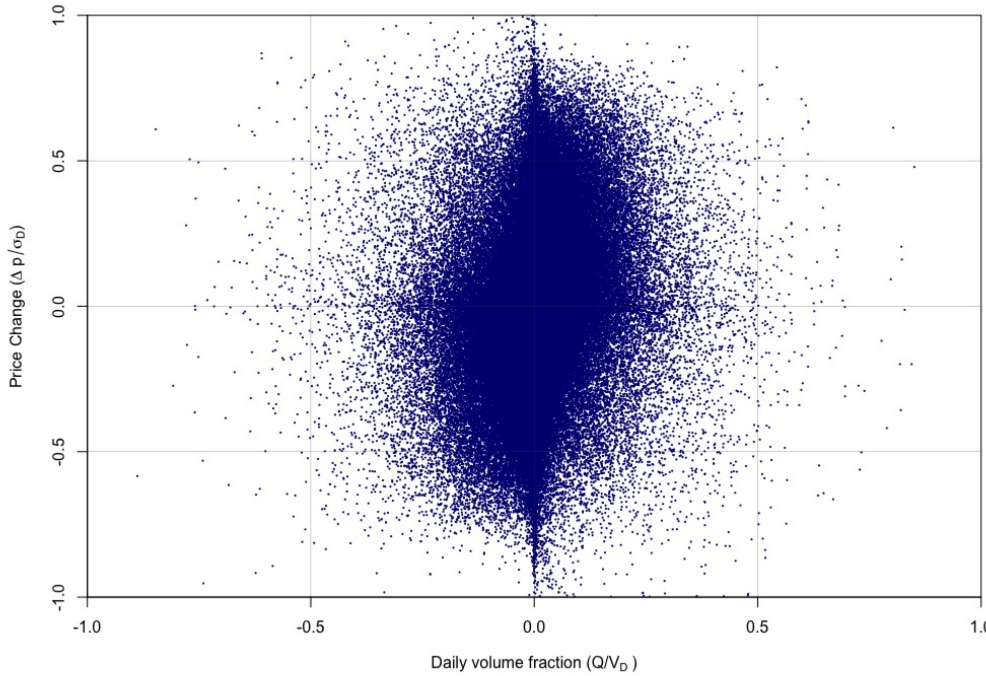

Fig. 2. Scatter-plot of price change ($\Delta p/\sigma_D$) versus daily volume fraction (Q/V_D) for all trades in our data (2008-2012).

four quadrants. In particular, the joint probabilities for each quadrant are as follows:

$$p(\Delta p > 0, Q > 0) = 33.0\%, \tag{9}$$

$$p(\Delta p < 0, Q > 0) = 19.0\%, \tag{10}$$

$$p(\Delta p > 0, Q < 0) = 17.7\%, \tag{11}$$

$$p(\Delta p < 0, Q < 0) = 30.3\%. \tag{12}$$

The conditional probabilities for a buy trade are:

$$p(\Delta p > 0 | Q > 0) = 63.5\%, \qquad p(\Delta p < 0 | Q > 0) = 36.5\%, \tag{13}$$

and for a sell trade:

$$p(\Delta p < 0 | Q < 0) = 63.1\%, \qquad p(\Delta p > 0 | Q < 0) = 36.9.\% \tag{14}$$

The conditional probabilities of equations (13) and (14) clearly show that the primary assumption of traditional impact models does not hold in our dataset. The sign of price changes does not seem to be determined by, or even significantly correlated with, the direction of the individual trade. In fact, we observe a *negative impact* in nearly 40% of the cases. This finding brings into question the received notion of 'price impact of a trade', that is, the idea that the price change contemporaneous to a trade is driven by that trade.

In market impact models that assume prices change with the direction of the trade, only noise can explain why prices move in the opposite direction of the trade. For instance, according to the square-root model of market impact, the price change between the arrival time and time of last fill is a square-root function of the daily volume fraction:

$$\Delta p = \varepsilon Y \sigma_D \sqrt{\frac{Q}{V_D}}, \tag{15}$$

where $\varepsilon = 1$ for buy trades and $\varepsilon = -1$ for sell trades, and Y is a constant. Adding a noise term we run the following regression:

$$\Delta p = Y\left(\varepsilon \sigma_D \sqrt{\frac{Q}{V_D}}\right) + \xi \tag{16}$$

which yields an estimated regression coefficient $\hat{Y} = 0.31$ in equation (16) with an estimated standard error of 0.002, a t-statistic of 180 and significance at the 1% level. Most importantly, the coefficient of determination is only $R^2 = 5.5\%$, which shows that the square-root model of market impact provides a poor explanation for the direction and magnitude of the returns. 94.5% of the variation in price change is given by the residual, whose standard deviation $\sigma_\xi \sim 0.01$ is of the same order as the daily volatility of the stock return, showing that volatility risk is by far the main determinant of the direction and magnitude of price variations during trade execution.

Given the trade direction, we can also use the square-root model to compute the conditional probabilities of points falling in the 'wrong' quadrants (where price change and trade direction have opposite signs) using the following logit model:

$$\begin{aligned} p\Big(\Delta p > 0 \Big| Q > 0\Big) &= \frac{1}{1 + exp\Big(-\beta\Big(\frac{Q}{V_D}\Big)^{\frac{1}{2}}\Big)}, \qquad (17) \\ p\Big(\Delta p < 0 \Big| Q > 0\Big) &= 1 - p\Big(\Delta p > 0 \Big| Q > 0\Big), \qquad (18) \end{aligned}$$

where Δp is the price change, $Q > 0$ for buys and $Q < 0$ for sells. We neglect the probability of a price change exactly equal to zero because this occurs in only 0.27% of our observations. The conditional probability of a negative price change from a buy trade in our data is 36.5%, while the estimated conditional probability from the model in equations 17-18 is only 7.4%, with a 99% confidence interval of $[7.0\% - 7.8\%]$. The empirical probability of a negative price change given a buy trade is therefore 70.5 standard errors away from what is predicted by the square-root model, with a p-value smaller than 0.00001. Similarly, the conditional probability of a positive price change from a sell trade is 36.9%, while the estimated conditional probability from the model in equations 17-18 is just 9.5%, with a 99% confidence interval of $[9.0\% - 10.0\%]$. The probability of a positive change given a sell trade is therefore 57.2 standard errors away from what is predicted by the square-root model, with a p-value smaller than 0.00001. These findings further support the view that the square-root model is inconsistent with the observed data and that the direction of price moves is not dominated by the buy/sell nature of individual trades.

3.2. Price changes and order flow imbalance

The results in subsection 3.1 undoubtedly show that the sign of the price change does not seem to be determined by, or to be significantly correlated with, the direction of the individual trade. Cont et al. (2014) demonstrate that price changes are well explained by a linear model driven by the *aggregate order flow*

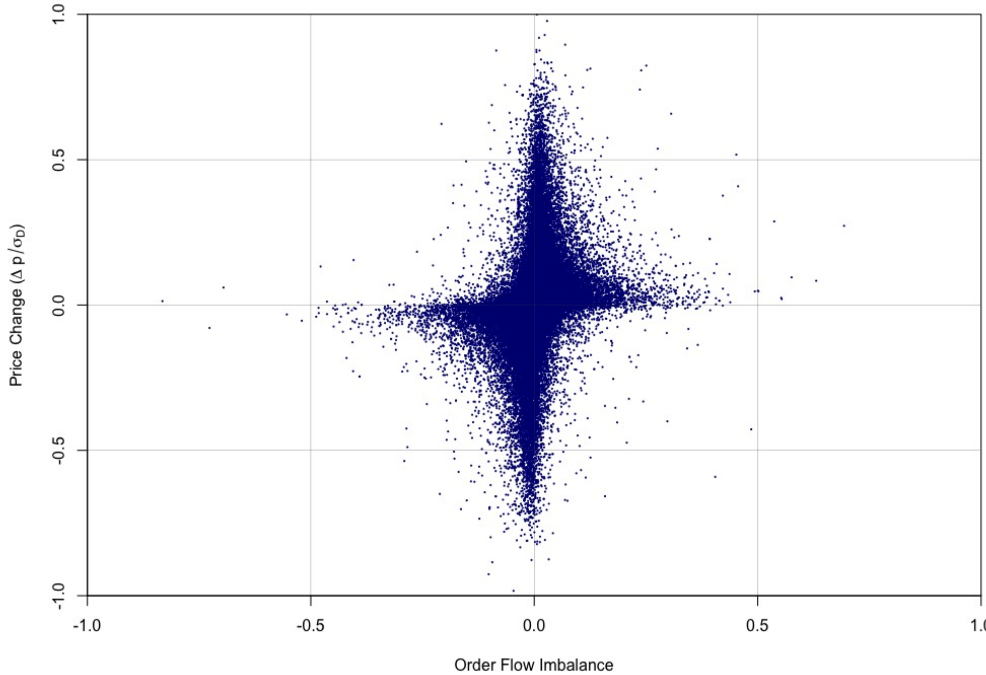

Fig. 3. Scatter-plot of price change ($\Delta p/\sigma_D$) versus order flow imbalance for all ANcerno trades in 77 selected Nasdaq-100 stocks (2008-2012).

imbalance (OFI). Following Cont et al. (2014), we evaluate this relationship between order flow imbalance and price changes on a subset of our ANcerno sample. In particular, we select all trades in 77 stocks that overlap the Nasdaq 100 index. This is because we can only compute OFI for Nasdaq stocks, for which the LOBSTER database records every single order book event at, at least, millisecond precision (Bibinger et al., 2017).

The scatter-plot in Figure 3 compares price changes versus OFI for this subset of ANcerno trades. The difference with Figure 2 is obvious. In particular, the joint probabilities are as follows:

$$p(\Delta p > 0, OFI > 0) = 36.6\%, \tag{19}$$

$$p(\Delta p < 0, OFI > 0) = 14.2\%, \tag{20}$$

$$p(\Delta p > 0, OFI < 0) = 13.7\%, \tag{21}$$

$$p(\Delta p < 0, OFI < 0) = 35.5\%. \tag{22}$$

which corresponds to the following conditional probabilities:

$$p(\Delta p > 0 | OFI > 0) = 70.1\%, \qquad p(\Delta p > 0 | OFI < 0) = 29.9\%, \tag{23}$$

$$p(\Delta p < 0 | OFI < 0) = 71.2\%, \qquad p(\Delta p > 0 | OFI < 0) = 28.8.\% \tag{24}$$

These conditional probabilities show that the sign of the price change is strongly correlated with the sign of the OFI, with a positive OFI leading to a positive price change in 70.1% of the cases, and a negative OFI

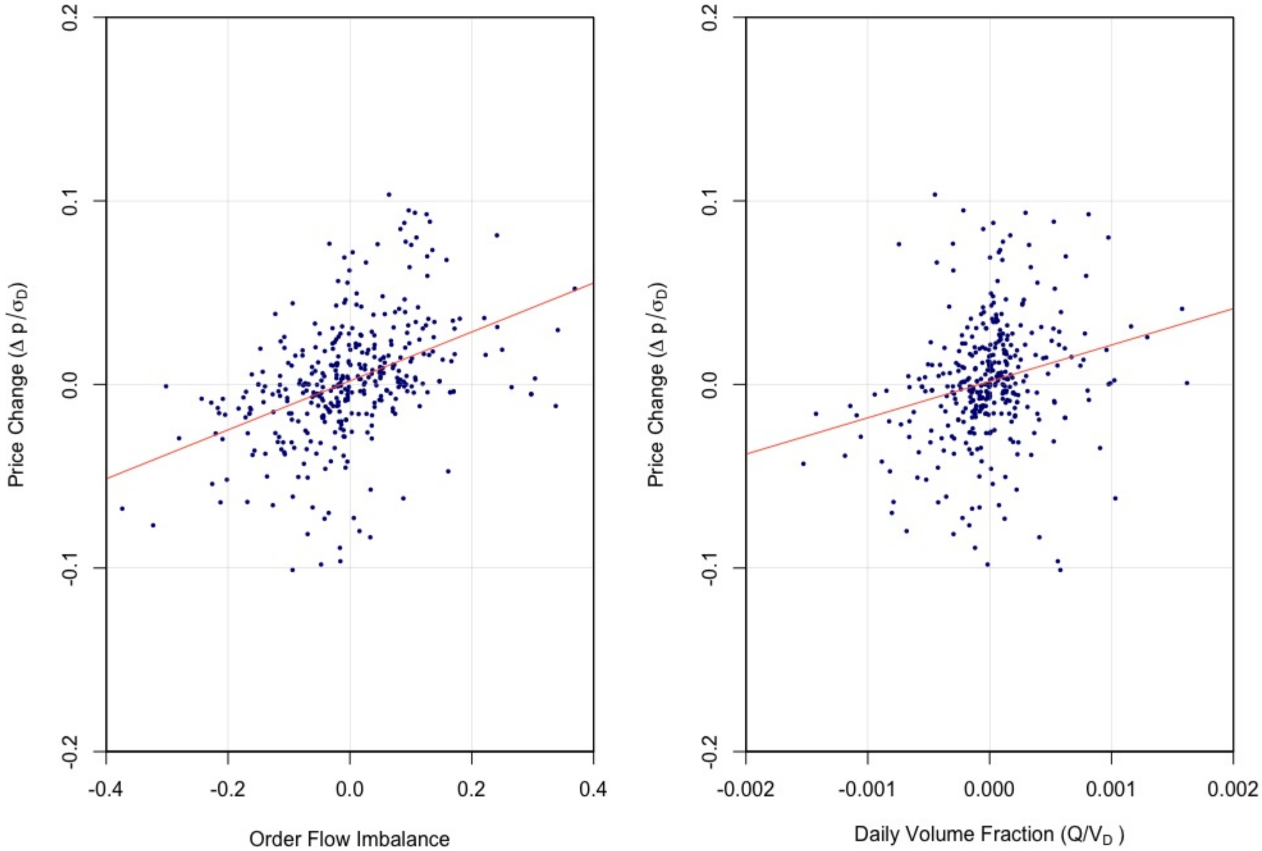

Fig. 4. Scatter-plots and linear regression lines of price change versus order flow imbalance (left) and price change versus daily volume fraction (right) for AAPL 2-minute trades (2008-2012). The estimated coefficients of determination are 18.6% (left) and 5.7% (right).

leading to a negative price change in 71.2% of the cases.

Following Cont et al. (2014), we run the following linear regression of price change on OFI:

$$\Delta p_{ik} = \alpha_{i,k} + \beta_{i,k} OFI_{i,k} + \xi_{i,k} \tag{25}$$

for each stock i and duration $k \in [2, 390]$ in our dataset, where the duration in expressed in minutes. $\xi_{i,k}$ is a noise term summarising influences of other factors (e.g., deeper levels of the order book).

Figure 4 shows a scatter-plot of price change versus OFI (left panel) and price change versus daily volume fraction (right panel) for all ANcerno trades in Apple (AAPL) with a duration of 2 minutes. A strong linear dependence of price change on OFI is exhibited, with a regression coefficient $\hat{\beta}_{i,k}$ which is statistically significant at the 1% level, and a coefficient of determination of 18.6%. In contrast, the coefficient of determination in the regression of price change on daily volume fraction is only 5.7%. We use heteroskedasticity-consistent Newey-West standard errors to compute our t-statistics, and find that the estimated regression coefficient,

$\hat{\beta}_{i,k}$, is statistically significant at the 5% level in 63.77% of the regressions, and at the 10% level in 72.20% of the regressions. The estimated intercept, or conditional mean, $\hat{\alpha}_{i,k}$, is significant at the 5% level in 6.29% of samples, and at the 10% level in 14.09% of samples. The average (absolute) t-statistics for $\hat{\alpha}_{i,k}$ and $\hat{\beta}_{i,k}$ in the cross-section are 34.94% and 18.28%, respectively; and the average coefficient of determination is 33.35%, with a standard deviation of 21.80.

These results demonstrate that the OFI provides a better explanation for the sign of price changes when compared to the direction of the individual trade. In other words, the price change is driven by the aggregate excess supply/demand rather than by the 'impact' of trades. These findings are consistent with a recent work on market impact in the US Treasury market, which finds that the order flow imbalance is the most important feature in predicting price changes (Salem et al., 2018, exhibit 5, p.7). They are also consistent with the findings of Easley et al. (2015), which shows that the trade side and order imbalance are key variables in modelling execution costs, and with the work of Bechler and Ludkovski (2015) which proposes a dynamic model combining the Almgren-Chriss framework, the order flow and the model of Easley et al. (2015). Overall, the results in this section invalidate the standard notion of the 'price impact of a trade' and imply that investors should focus on aggregate order flow modelling, rather than trade-induced price moves.

4. The amplitude of price changes during trade execution

4.1. Where does the square-root law come from?

The previous section demonstrated that it is the aggregate (net) order flow imbalance, rather than the individual trade, that determines the sign of price changes and therefore that the idea that the 'impact' of a trade is the main determinant of price changes needs to be rethought. So, why do empirical studies consistently find a square-root dependence of 'impact' on trade size? The reason is that these empirical studies, like ours, are based on datasets composed of *orderly* institutional trade executions, defined as an execution in which the trade *duration* T is modulated according to the trade size Q in order to maintain a participation rate below some reasonable bound (Avellaneda and Cont, 2013). In fact, the two variables are *proportional* if trading occurs at a fixed participation rate throughout the day. If $\overline{\pi}$ is the average participation rate, then

$$Q = \overline{\pi}\, V_D T(Q). \tag{26}$$

When returns are uncorrelated, the amplitude of a price move over a horizon τ scales like $\sigma\sqrt{\tau}$. So, over the trade duration T, the typical amplitude of a price change will be:

$$\sigma\sqrt{T} = \sigma\sqrt{\frac{Q}{\overline{\pi} V_D}} = Y\sigma\sqrt{\frac{Q}{V_D}} \tag{27}$$

where $Y = 1/\sqrt{\overline{\pi}}$. In other words, the 'square-root law' of market impact turns out to be nothing more than the (well known) square-root scaling of volatility with the execution horizon (Bachelier, 1900), which explains the ubiquity of this 'law'.

Our explanation is consistent with previous findings on the influence of trade duration on market impact. Engle et al. (2012) and Easley et al. (2015) emphasised the temporal dimension of transaction cost analysis and the importance of the execution horizon in the analysis of execution costs. Bacry et al. (2015) present a market impact model where the return is a function of the daily volume fraction and duration (Bacry et al., 2015, eqn. 4, and Table 2) with power-law dependence with exponents 0.54-0.59 for the daily volume

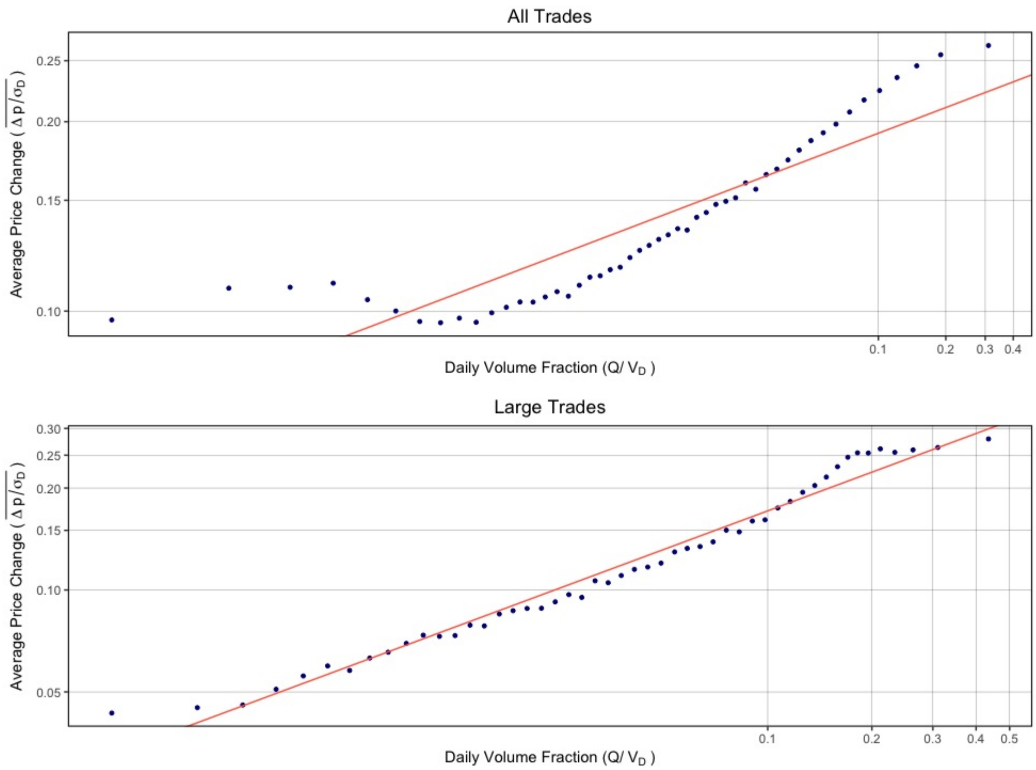

Fig. 5. Scatter-plots and linear regression lines of (log) *Average* price change $(\overline{\Delta p/\sigma_D})_n$ versus (log) daily volume fraction $(Q/V_D)_n$, equation (29), for all trades (upper panel) and large trades only (lower panel) (2008-2012). The estimated regression coefficients are 0.107 for all trades and 0.381 for large trades.

fraction and between -0.35 and -0.23 for the duration but not perform any analysis of impact conditional on duration. Bershova and Rakhlin (2013) analyse trades executed by Alliance Bernstein's trading desk and run a non-linear regression of market impact on trade duration for *all* observations, without averaging. They find that impact is a square-root function of *duration* and report a coefficient of determination of 29% (Bershova and Rakhlin, 2013, eqn. 7, Table 5 and Fig. 4). Finally, Salem et al. (2018) examine a data set of trade execution in the US Treasury market and report that price impact increases with duration (Salem et al., 2018, p. 9).

4.2. Conditioning on trade duration

The explanation for the 'square-root law' findings in the literature provided in (27) is empirically testable. If the square-root law arises from the square-root scaling of volatility with trade duration, we should find that price changes during execution do not depend on trade size when conditioned on duration. Before we test this hypothesis, we show in Figure 5 the *average* 'market impact' curves, as they are usually presented in the square-root law literature (Zarinelli et al., 2015): the daily volume fraction (Q/V_D) is divided into $N = 50$ equally-populated bins, and then the conditional expectation of the price change is computed for

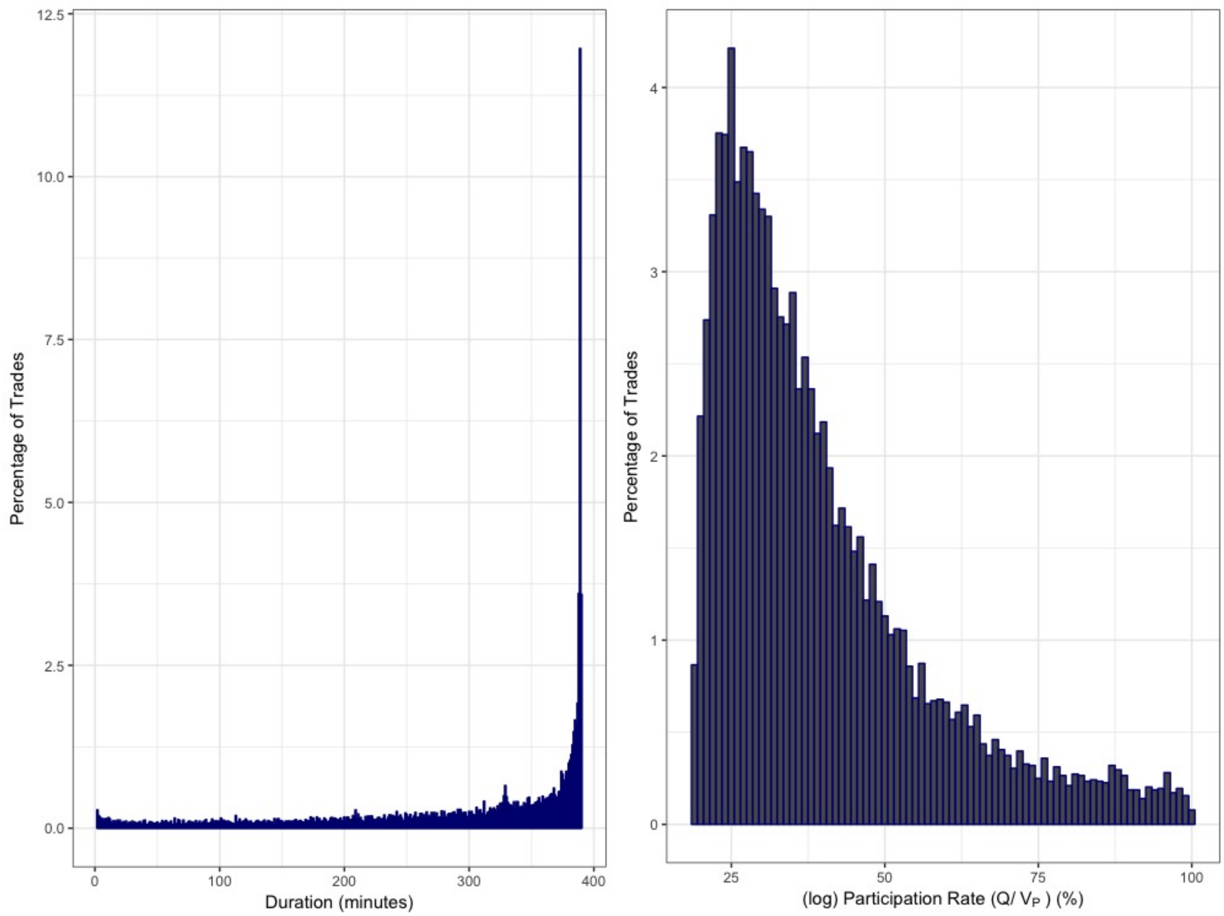

Fig. 6. Histogram of trade duration values (left) and (log) participation rate for all trades with a daily volume fraction (Q/V_D) above 18.5% (2008-2012).

each bin. More precisely, the average price change for bin $n \in [1, N]$ is computed as:

$$\left(\overline{\frac{\Delta p}{\sigma_D}}\right)_n = \frac{1}{J}\sum_{j=1}^{J} \varepsilon\left(\frac{\Delta p_j}{\sigma_{D,j}} \,\Big|\, \frac{Q}{V_D} \in n\right), \tag{28}$$

where $j = 1, ..., J$ are the trades in bin n, and $\varepsilon = \pm 1$ is the trade direction. The average 'impact' curve for all trades in our dataset is presented in the upper panel and the curve for large trades only is presented in the lower panel. If we run the following logarithmic regressions:

$$\log\left(\overline{\frac{\Delta p}{\sigma_D}}\right)_n = \alpha_n + \beta_{s,n} \log\left(\overline{\frac{Q}{V_D}}\right)_n + \xi_n, \tag{29}$$

where $\overline{Q/V_D}$ is the average daily volume fraction for bin n, we find that the square-root law model holds reasonably well for large trades, where the estimated regression coefficient $\hat{\beta}_{s,n}$ is 0.381, but fails to explain all trades, where $\hat{\beta}_{s,n}$ is only 0.107. Further, it is clear from the upper panel, that both small and very large trades deviate from the square-root law. In particular, the lower panel shows that the average price change is essentially the same for all trades with a daily volume fraction of 18.5% and above. In Figure 6, we show the histograms of the duration values and (log) participation rate for these trades. It is clear from the left panel of Figure 6 that these are very large trades which are mostly executed over the duration of the whole trading day. It is then not surprising to see these that the price change associated with these VWAP trades is the same regardless of the quantity traded, because this is nothing else than the daily return.

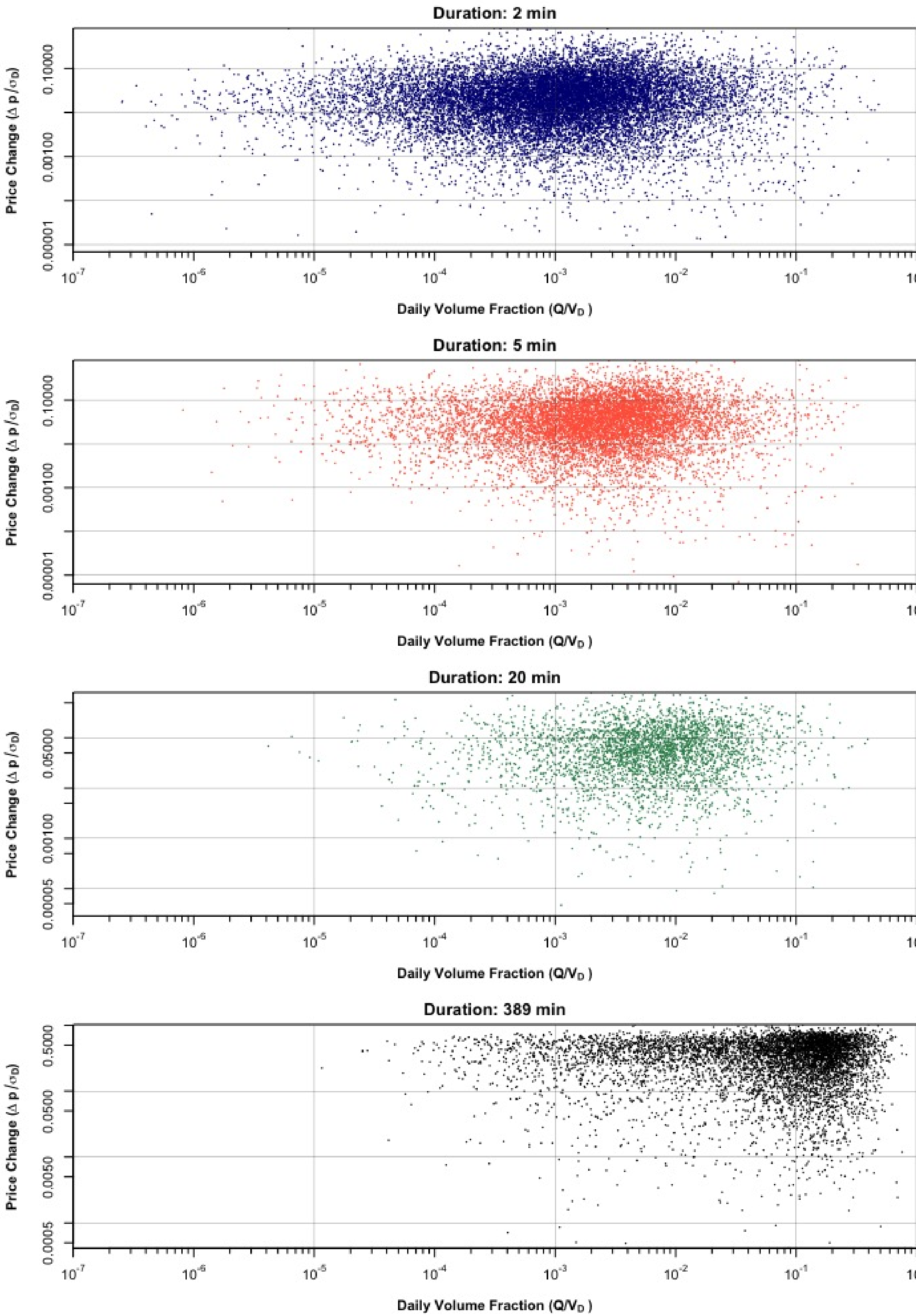

Fig. 7. Scatter-plots of (log) price change ($\Delta p/\sigma_D$) versus (log) daily volume fraction (Q/V_D) for selected fixed duration values (2008-2012).

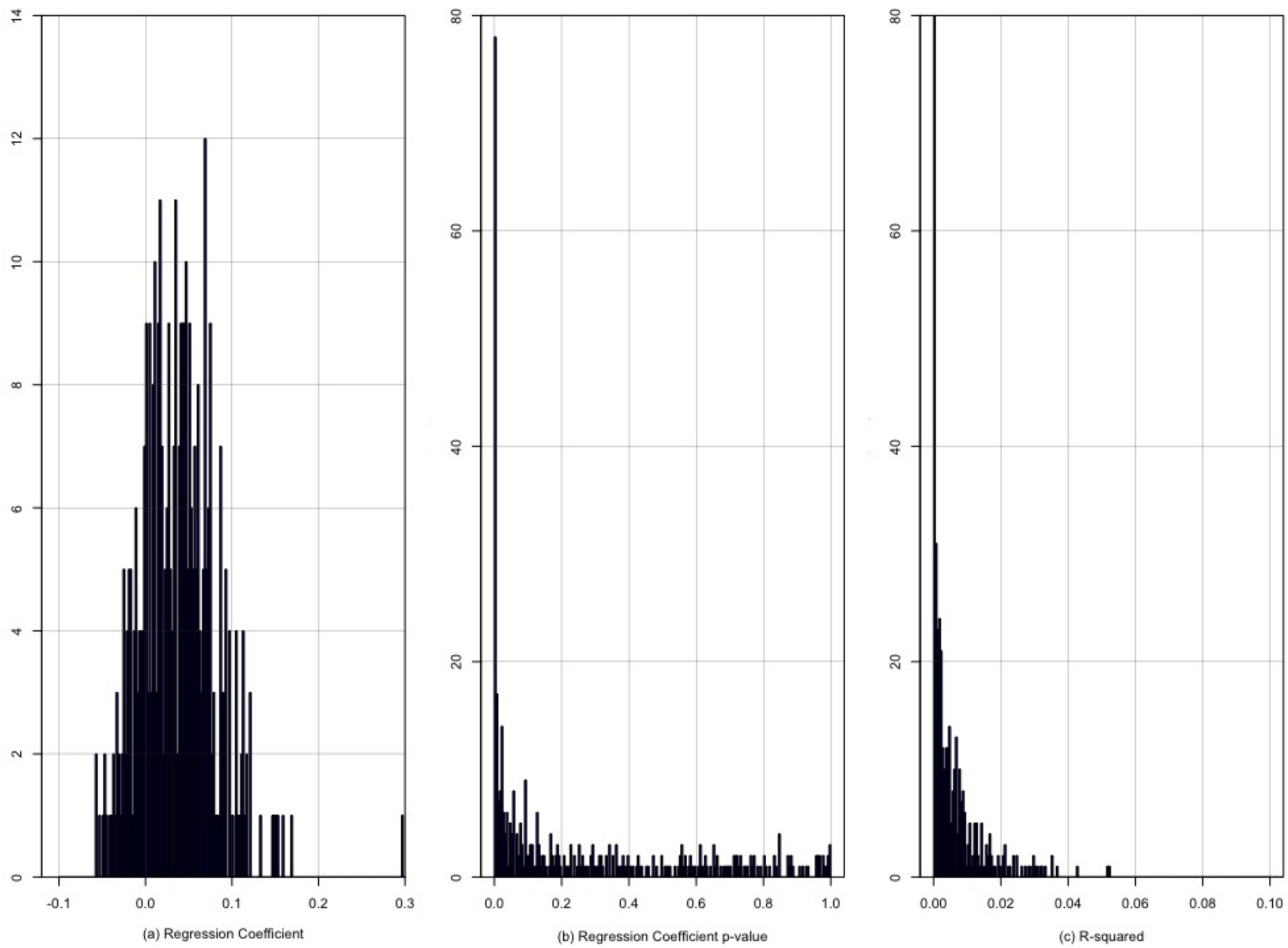

Fig. 8. Histograms of estimated regression coefficients ($\beta_{s,k}$) (a), p-values for the regression coefficients (b), and R-squared from the regression (c) of $\log|\Delta p_k/\sigma_D| = \alpha_k + \beta_{s,k} \log Q/V_D + \xi_k$, equation(30), conditional on each of the $k \in [2, 390]$ duration values for all trades (2008-2012).

If we look at all trades and condition on trade duration, the relationship between price changes and trade size disappears. This is clear from Figure 7, which shows four scatter-plots of (log) price changes ($\Delta p/\sigma_D$) versus (log) daily volume fraction (Q/V_D) conditioned on representative short- (2 minutes), medium- (5 minutes), and high-duration (389 minutes) values. The cloud of points for each of these four duration values is essentially flat, showing no relationship between price changes and the trade size. To test this formally, we run the following regression:

$$\log\left|\frac{\Delta p_k}{\sigma_D}\right| = \alpha_k + \beta_{s,k} \log \frac{Q}{V_D} + \xi_k, \tag{30}$$

for each duration $k \in [2, 390]$ in our dataset, where the duration in expressed in minutes. Histograms of the regression coefficient $\hat{\beta}_{s,k}$, its p-value and the coefficient of determination for all 389 durations are presented in Figure 8. $\hat{\beta}_{s,k}$ is statistically significant at the 5% level in 146 of the 389 durations (37.5%). However,

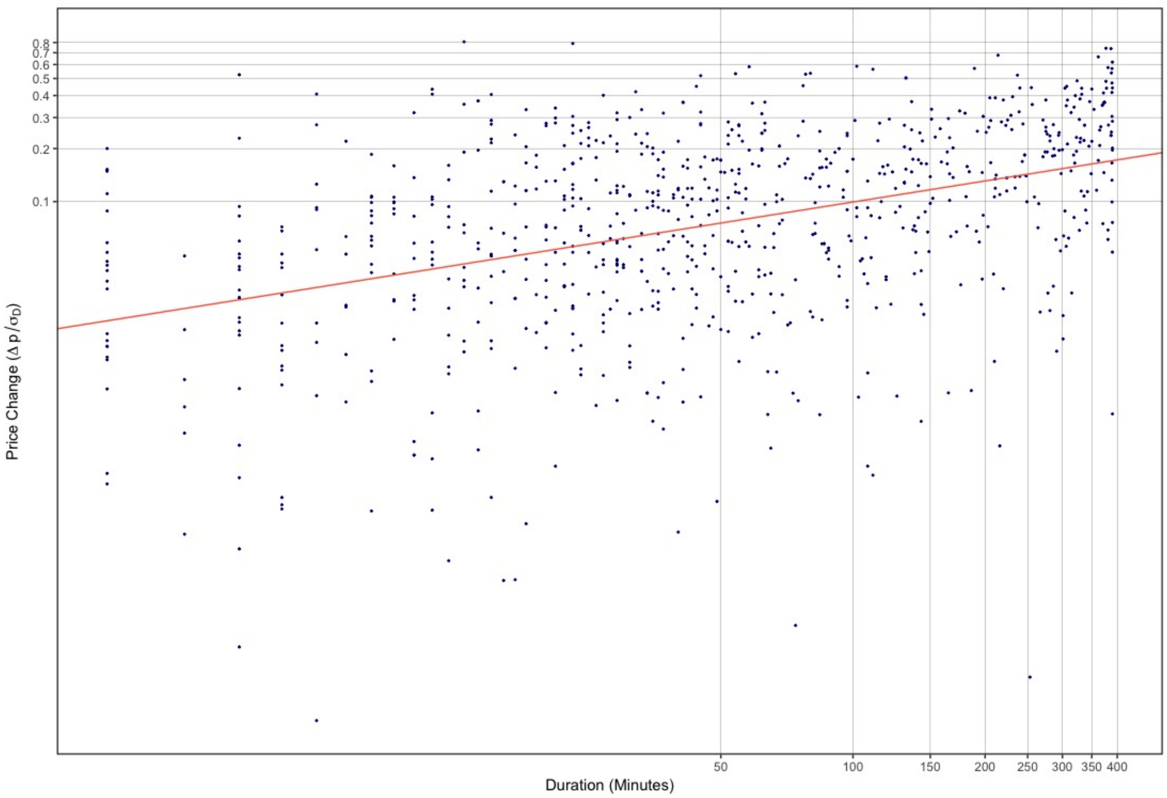

Fig. 9. Scatter-plot and linear regression line of log price change versus log duration for fixed daily volume fraction of 1% for all trades (2008-2012), equation (31). The estimated regression coefficient ($\hat{\beta}_{d,n}$) is 0.40 and the coefficient of determination (R^2) is 16.5%.

in the vast majority of cases, its estimate is very close to zero (panel (a)). On average, $\hat{\beta}$ is 0.04, with an average coefficient of determination of only 0.7%, with 358 out of 389 (92%) regressions having an R^2 smaller than 2%. In other words, once conditioned on duration, trade size has nearly zero explanatory power on the price change.

These results strongly support the explanation provided above for the 'square root law of market impact' as arising from the square-root scaling of volatility with trade duration (Bachelier, 1900). The implication is that the two components of execution costs –'market impact' and 'volatility risk'– are in fact two sides of the same coin: market volatility.

4.3. Conditioning on trade size

Our explanation for the square-root law as being a manifestation of the square root scaling of volatility with duration implies that 'market impact' –i.e. the expected amplitude of price changes during execution– increases with duration: if we increase the trade duration from T to kT then the impact should *increase* by a factor $\sqrt{k}$. This contradicts one of the properties of the square-root impact model (1) according to which the market impact of a trade of size Q does not depend on the trade duration, sometimes referred to as the "invariance" property of the square-root law model (Bouchaud et al., 2018).

Since our dataset contains a wide range of participation rates, we can test which of these two hypotheses

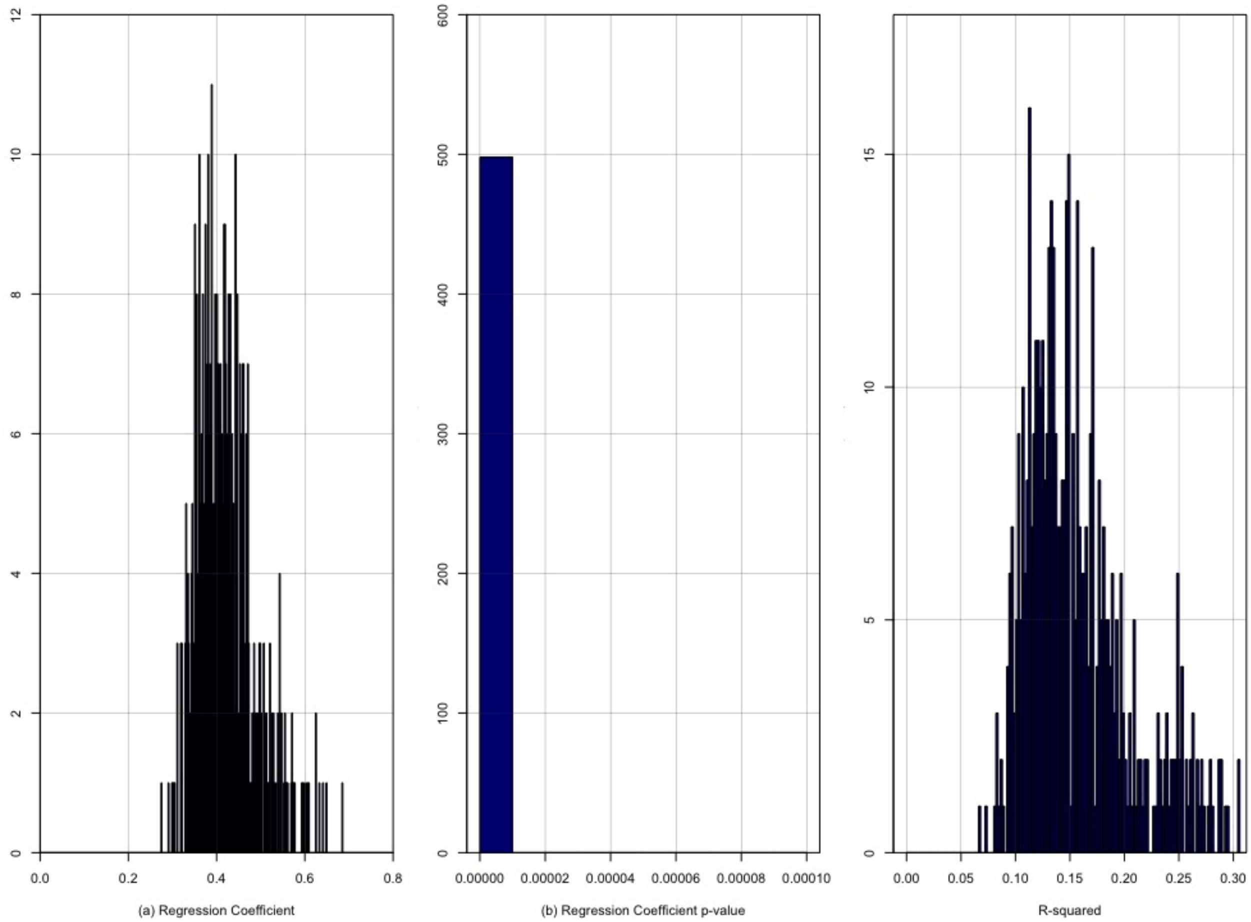

Fig. 10. Histograms of estimated regression coefficients ($\hat{\beta}_{d,n}$) (a), p-values for the regression coefficients (b), and R-squared (c) from the regressions $\log|\Delta p_n/\sigma_D| = \alpha_n + \beta_{d,n} \log T + \xi_n$ (equation (31)) conditioned on each of the $n \in [1, 500]$ daily volume fraction (Q/V_D) bins (2008-2012).

is correct by analysing the dependence of price changes on trade duration conditioned on trade size. First, the daily volume fraction (Q/V_D) is divided into 500 equally populated bins $n \in [1, 500]$. Then, for each bin a regression of price changes on duration is estimated according to,

$$\log\left|\frac{\Delta p_n}{\sigma_D}\right| = \alpha_n + \beta_{d,n} \log T + \xi_n. \tag{31}$$

Figure 9 shows a scatter-plot and regression for trades with daily volume fraction of 1%. The dependence of price change on duration is clearly exhibited when conditioned on trade size. In particular, the $\hat{\beta}_{d,n}$ coefficient is statistically different from zero at the 1% level, with a value of 0.40 and coefficient of determination of 16.5%. This supports the hypothesis that a square-root relationship exists between price change and duration, and not between price change and trade size.

Figure 10 shows the histograms for the $\hat{\beta}_{d,n}$ coefficient, its p-value and the respective coefficient of determination of regression (31) for all 500 conditional values of Q/V_D. The average $\hat{\beta}_{d,n}$ is 0.42 and statistically different from zero at the 1% level in all regressions. The average coefficient of determination is 15.7%.

These results demonstrate that during an orderly execution, the amplitude of price changes is determined by trade duration rather than trade size. Further, they demonstrate a dependence on duration and not an invariance. This finding is important because invariance suggests that the speed of execution is not important while the data shows that it is in fact the main determinant of the amplitude of price changes. This result is supported by, and consistent with other recent studies on the role of trade duration in equity markets Bershova and Rakhlin (2013) and bond markets (Salem et al., 2018).

4.4. Joint analysis

A step-wise multivariate log-regression of the absolute price on trade size, duration and volatility further supports our results. We proceed as follows. First, we perform a log-log regression of price changes on the daily volume fraction (Q/V_D). Then, a log-log regression of price changes on trade duration (T). Finally, a regression with both explanatory variables. In each case, the daily volatility (σ_D) is included as an additional explanatory variable. The three regressions are as follows:

$$\log|\Delta p| = \alpha + \beta_s \log\frac{Q}{V_D} + \beta_v \log\sigma_D + \xi, \tag{32}$$

$$\log|\Delta p| = \alpha + \beta_d \log T + \beta_v \log\sigma_D + \xi, \tag{33}$$

$$\log|\Delta p| = \alpha + \beta_s \log\frac{Q}{V_D} + \beta_d \log T + \beta_v \log\sigma_D + \xi. \tag{34}$$

The results of these regressions are presented in Table 2 for all trades (columns 2-4) and only large trades (columns 5-7). In the first regression for all trades (column 2), the regression coefficient on the daily volume fraction, $\hat{\beta}_s$, is 0.201, and is statistically significant at the 1% level. This is very far from 1/2, so we do not find any evidence to support the 'square-root law' in terms of trade size.

In the second regression for all trades (column 3), the regression coefficient on trade duration, $\hat{\beta}_d$, is 0.437 and is statistically significant at the 1% level. This is much closer to 1/2, and supports our previous finding that price changes are a square-root function of trade *duration* rather than trade size. Also, as expected, we find a regression coefficient close to one on the daily volatility ($\hat{\beta}_v = 0.911$), and we notice that the coefficient of determination for this regression is much higher than the first regression (34.3% versus 21.2%).

Finally, when daily volume fraction and duration are included in the regression (column 4), the coefficient on trade duration remains close to 1/2 (specifically 0.414) while the regression coefficient on daily volume fraction drops to only 0.038. Both regression coefficients are statistically significant at the 1% level and collinearity between the two variables is rejected based on the variance inflation factor test.[5] The inclusion of the daily volume fraction as a statistically significant explanatory variable is confirmed when we compare the nested models in equations (13) and (14), with an ANOVA test (F- statistic: 1097.5, p-value: $< 2.2\,10^{-16}$) or using the Akaike Information Criterion (1,355,981 versus 1,357,075). However, despite being statistically significant, the estimated regression coefficient on the daily volume fraction is very small and it implies that a 1% increase in daily volume fraction leads to only a 0.038% increase in price change. Moreover, the increase in the coefficient of determination compared to the second regression is minimal: from 34.3% to 34.5%.

[5] The square root of the variance inflation factor indicates the degree to which the confidence interval for the estimated regression coefficient is expanded relative to a model with uncorrelated predictors (Fox and Monette, 1992). In this case we find values of 1.15 for both variables, suggesting that multicollinearity is not a problem.

Table 2: Estimated log-log regressions of price change on daily volume fraction, duration and daily volatility for all trades (left) and only for large trades (right) (2008-2012), equations (32 - 34). In parentheses: heteroskedasticty-consistent Newey-West standard errors.

	Dependent variable: Price Change (Δp)					
	All Trades			Large Trades		
Model	(32)	(33)	(34)	(32)	(33)	(34)
Constant α	-2.071^{***}	-2.070^{***}	-1.954^{***}	-1.675^{***}	-2.151^{***}	-2.033^{***}
	(0.013)	(0.012)	(0.014)	(0.027)	(0.026)	(0.040)
Trade size Q/V_D	0.201^{***}		0.038^{***}	0.451^{***}		0.071^{***}
	(0.001)		(0.001)	(0.003)		(0.005)
Duration T		0.437^{***}	0.414^{***}		0.464^{***}	0.412^{***}
		(0.001)	(0.001)		(0.003)	(0.005)
Volatility σ_D	0.895^{***}	0.911^{***}	0.905^{***}	0.888^{***}	0.874^{***}	0.874^{***}
	(0.003)	(0.003)	(0.003)	(0.007)	(0.006)	(0.007)
Observations	410,792	410,792	410,792	102,698	102,698	102,698
R^2	0.212	0.343	0.345	0.284	0.335	0.346
Adjusted R^2	0.212	0.343	0.345	0.284	0.335	0.336
Residual Std. Error	1.382	1.262	1.260	1.350	1.301	1.300
F Statistic	$55{,}307^{***}$	$107{,}191^{***}$	$72{,}017^{***}$	$20{,}361^{***}$	$25{,}892^{***}$	$17{,}357^{***}$
AIC	1,431,635	1,357,075	1,355,981	353,163	345,533	345,343

Note: *p<0.1; **p<0.05; ***p<0.01

The findings are similar if we focus only on large trades. In the first regression (column 5), the regression coefficient on the daily volume fraction, $\hat{\beta}_s$, is 0.451, and is statistically significant at the 1% level. This is in line with the stylised fact of a 'square-root law' in terms of trade size reported in the literature. However, if we look at the results of the second regression (column 6), the estimated regression coefficient on trade duration $\hat{\beta}_d$, is also close to 1/2 (0.464) and statistically significant at the 1% level, and the coefficient of determination for this regression is higher (33.5% versus 28.4%).

Finally, when daily volume fraction and duration are included in the regression (column 7), the coefficient on trade duration remains close to 1/2 (specifically 0.412) while the regression coefficient on daily volume fraction drops to only 0.071. Both regression coefficients are statistically significant at the 1% level, and the inclusion of the daily volume fraction as a statistically significant explanatory variable is confirmed when we compare the nested models in equations (13) and (14), with an ANOVA test (F- statistic: 191.4, p-value: $< 2.2\ 10^{-16}$) or using the Akaike Information Criterion (345,343 versus 345,533). However, despite being statistically significant, the estimated regression coefficient on the daily volume fraction is very small and it implies that a 1% increase in daily volume fraction leads to only a 0.071% increase in the price change.

For large trades, we also notice that that the sum of the estimated coefficients on duration and daily volume fraction when both variables are included is similar to the coefficient on trade duration alone, when only this variable and daily volatility are considered; that the standard error on $\hat{\beta}_d$ doubles when we include daily volume fraction; and that the increase in the coefficient of determination is small: from 33.5% to 34.6%. Suspecting a multicolinearity problem in equation (34), we compute the variance inflation factor, whose square root indicates the degree to which the confidence interval for the estimated regression coefficient is expanded relative to a model with uncorrelated predictors. We find a value of around 1.779 on both daily volume fraction and duration, and one on daily volatility. These results confirm that the two variables are collinear, and suggest that for large trades the dependence between price change and daily volume fraction mainly comes from the correlation between trade size and duration, which is 0.83. This is not surprising, since we are looking at very large trades, whose participation rate is greater than 16%. These trades are likely to have been executed over similarly long durations.

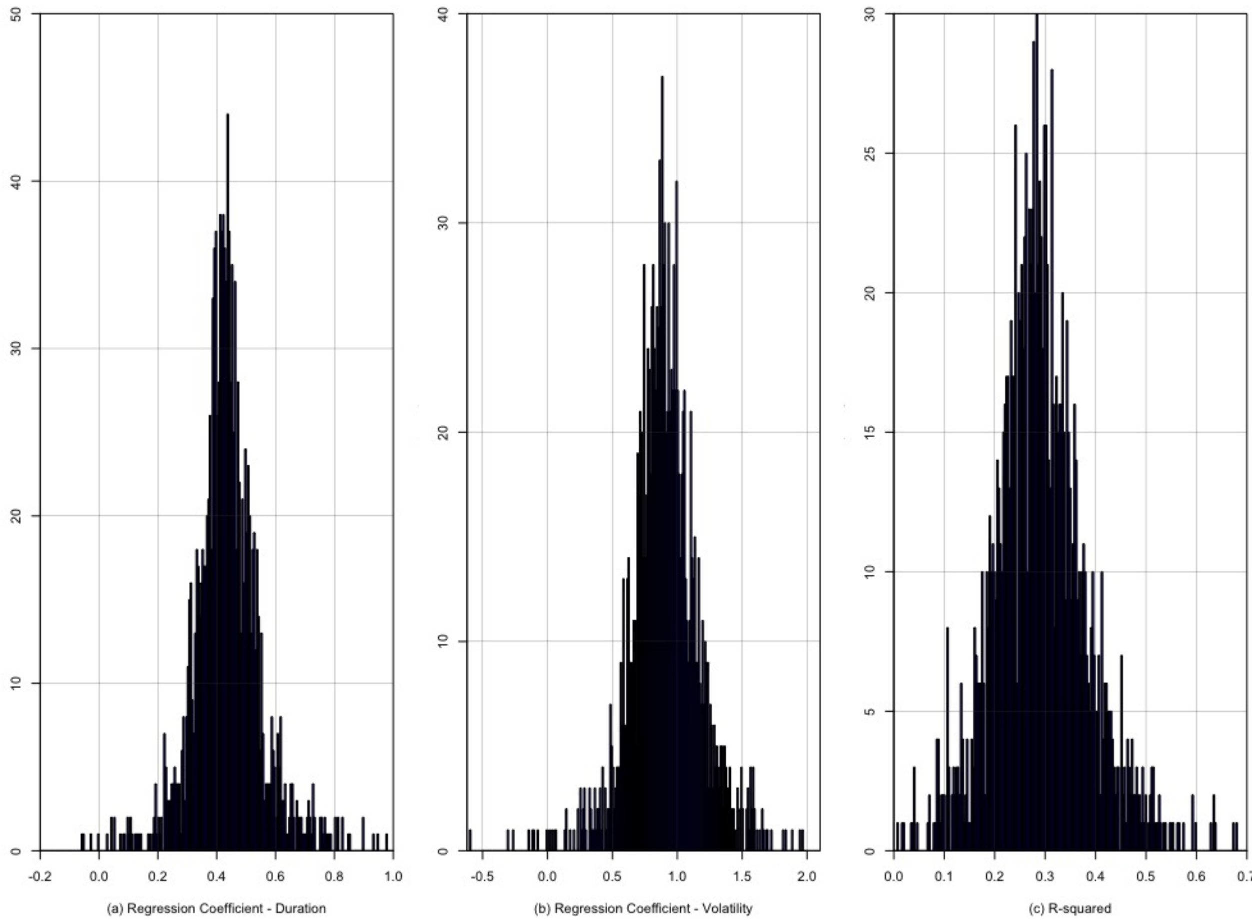

Fig. 11. Histograms of estimated regression coefficients for duration ($\hat{\beta_d}$) (a), and volatility ($\hat{\beta_v}$) (b), and R-squared (c) from the regression $\log|\Delta p| = \alpha + \beta_d \log T + \beta_v \log \sigma_D + \xi$ (Eq. (33)) for each stock in our database (2008-2012).

The model in equation (33) is also independently estimated for each stock. Figure 11 shows the histograms of the estimated regression coefficients and the coefficients of determination. The mean value of $\hat{\beta}_d$ is 0.448, and we can see how most of the estimated $\hat{\beta}_d$ coefficients are indeed close to 1/2. In particular, 95.51% of $\hat{\beta}_d$ coefficients are statistically significant at the 5% level. The mean value of $\hat{\beta}_v$ is 0.914, and we can see how most of the estimated $\hat{\beta}_v$ coefficients are indeed close to one, as we would expect. In particular, 94.07% of $\hat{\beta}_d$ coefficients are statistically significant at the 5% level. The average coefficient of determination is 28.98% and its maximum value is 68.68%.

The results from these multiple-regressions show very clearly that the variation of the amplitude of price changes accompanying trade executions is much better explained in terms of a square-root dependence on trade *duration* rather than trade size, with volatility being the proportionality coefficient. As explained in section 4.1, this square-root relationship arises naturally from well known square root scaling of volatility with horizon and does not require to invoke any concept of 'market impact'. These findings are in line with Bershova and Rakhlin (2013) and imply that rather than trying to optimise 'impact' at a trade-by-trade level, investors should focus on managing price volatility.

5. The role of trading speed and participation rate

Most optimal execution models consider the participation rate, which is a measure of the speed of trading, as the key driver of market impact. The assumption made by these models is that faster trading leads to higher market impact and therefore to higher execution costs.

However, as discussed in Section 4.2, the analysis of large trades shows no such dependence of impact on the participation rate or the trading speed . As shown in Figure 5 (lower panel), for trades executed over the entire trading day with size larger then 18.5% of daily volume, an increase in the size, and therefore in the participation rate has essentially no influence on the amplitude of the price change during the trade.

Driven by these observations, we take a closer look at the influence of the (average) participation rate on price impact, conditional on trade duration. Figure 12 displays scatter-plots of (log) price changes ($\Delta p/\sigma_D$) versus (log) participation rate (Q/V_P) conditioned on representative short- (2 minutes), medium- (5 minutes), and high-duration (389 minutes) values. These plots are essentially flat, showing no relationship between price changes and the participation rate. To confirm this hypothesis, we run the following regression:

$$\log\left|\frac{\Delta p_k}{\sigma_D}\right| = \alpha_k + \beta_{p,k}\log\frac{Q}{V_P} + \xi_k, \tag{35}$$

for each duration $k \in [2, 390]$ in our dataset, where the duration in expressed in minutes. Histograms of the regression coefficient $\hat{\beta}_{p,k}$, its p-value and the coefficient of determination for all 389 durations are presented in Figure 13. $\hat{\beta}_{p,k}$ is statistically significant at the 5% level in 46 of the 389 durations (11.8%), and its average estimate is only 0.02 (panel (a)). The average coefficient of determination for the regression in (35) is only 0.5%, with 372 out of 389 (96%) regressions having an R^2 smaller than 2%. In other words, once conditioned on trade duration, the participation rate has essentially zero explanatory power on price changes. More importantly, we also find *negative* regression coefficients, implying that an increase in the participation rate during the same trade horizon, and therefore an increase in the trading speed, leads to *lower* market impact as in Salem et al. (2018). Thus, these results strongly contradict the notion that the participation rate is *the* main driver of execution costs, as emphasised in much of the current literature on optimal execution.

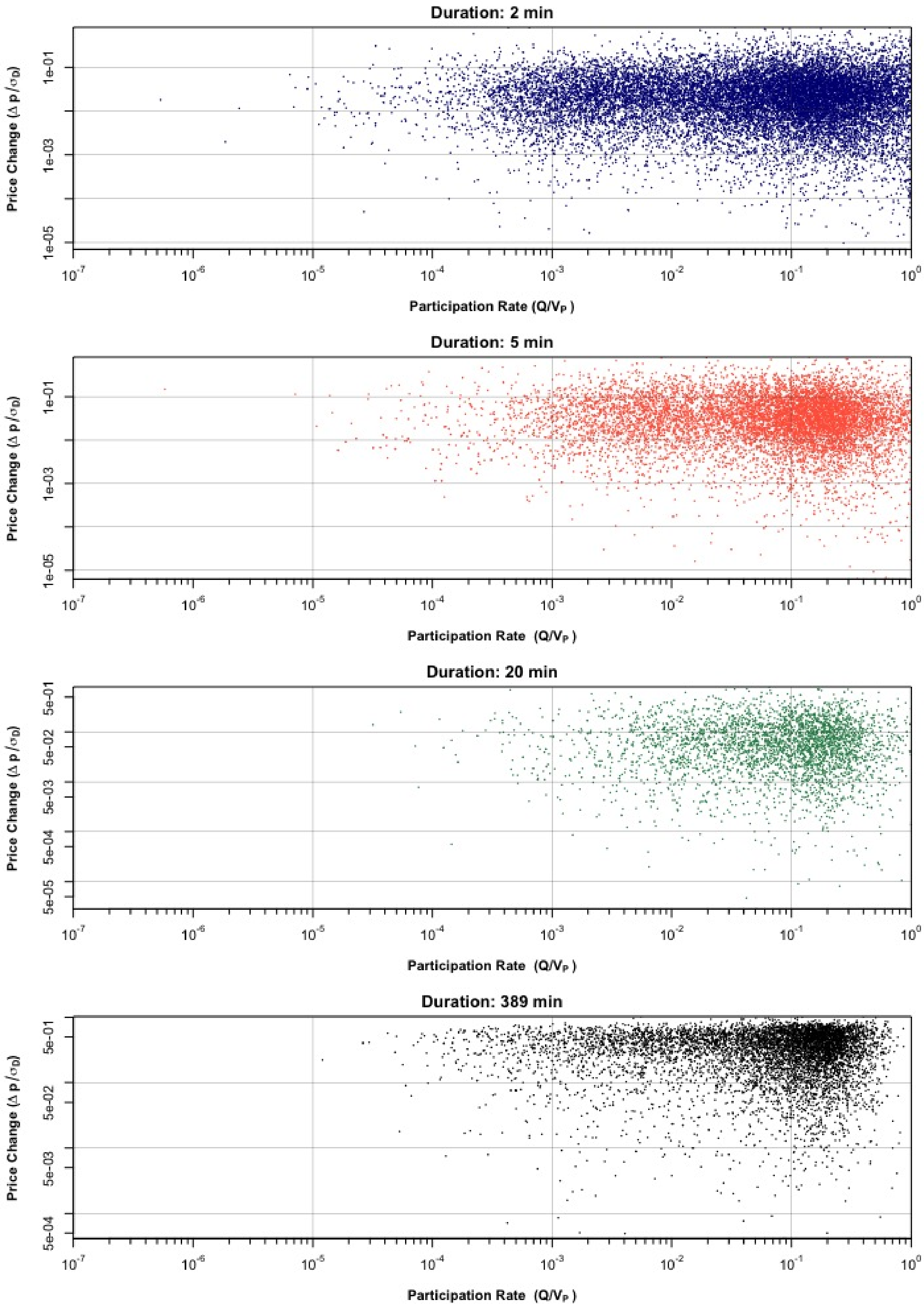

Fig. 12. Scatter-plots of (log) price change ($\Delta p/\sigma_D$) versus (log) participation rate (Q/V_P) for selected fixed duration values (2008-2012).

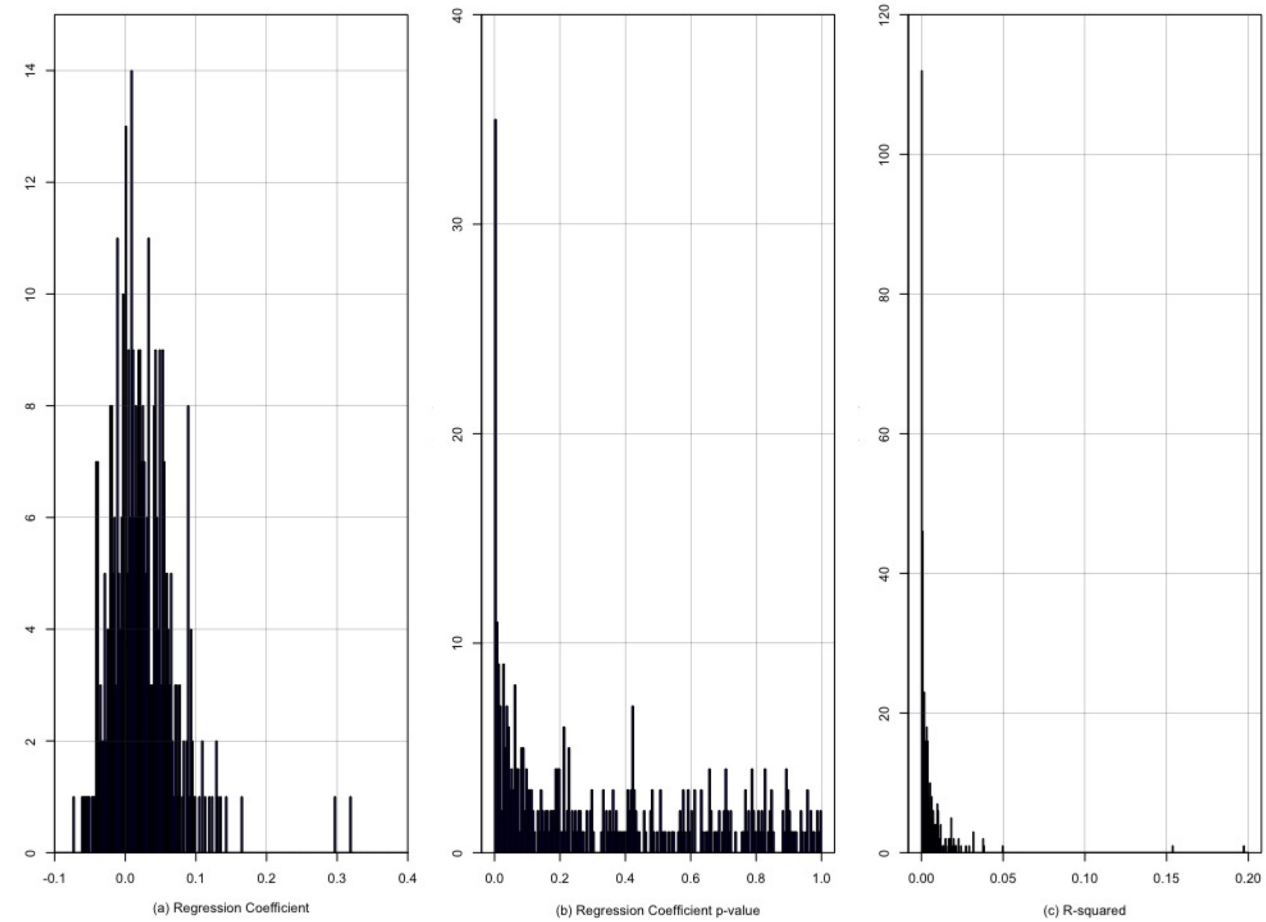

Fig. 13. Histograms of estimated regression coefficients $(\hat{\beta}_{p,k})$(a), p-values for the regression coefficients (b), and R-squared from the regression (c) of $\log|\Delta p_k/\sigma_D| = \alpha_k + \beta_{p,k}\log\frac{Q}{V_P} + \xi_k$, equation (35), conditional on each of the $k \in [2, 390]$ duration values for all trades (2008-2012).

We also run a step-wise multivariate log-regression of the absolute price on the participation rate, duration and volatility to further investigate the role of the participation rate on price changes, once trade duration is controlled for. We estimate the following log-log regressions:

$$\log|\Delta p| = \alpha + \beta_p \log\frac{Q}{V_P} + \beta_v \log\sigma_D + \xi, \tag{36}$$

$$\log|\Delta p| = \alpha + \beta_d \log T + \beta_v \log\sigma_D + \xi, \tag{37}$$

$$\log|\Delta p| = \alpha + \beta_p \log\frac{Q}{V_P} + \beta_d \log T + \beta_v \log\sigma_D + \xi. \tag{38}$$

The results of these regressions are presented in Table 3 for all trades (columns 2-4) and only large trades (columns 5-7). In the first regression for all trades (column 2), the estimated regression coefficient on the participation rate, $\hat{\beta}_p$, is -0.014 and is statistically significant at the 1% level. The *negative* regression coefficient is clearly at odds with the notion that higher participation rates lead to higher market impact.

Table 3: Estimated logarithmic regressions of price change on participation rate, duration and daily volatility for all trades (left) and only for large trades (right) equations (36 - 38) (2008-2012). In parentheses: heteroskedasticty-consistent Newey-West standard errors.

	Dependent variable: Price Change (Δp)					
	All Trades			Large Trades		
Model	(36)	(37)	(38)	(36)	(37)	(38)
Constant α	-2.949^{***}	-2.071^{***}	-2.039^{***}	-3.698^{***}	-2.151^{***}	-2.312^{***}
	(0.013)	(0.012)	(0.013)	(0.031)	(0.026)	(0.028)
Participation Rate (Q/V_P)	-0.014^{***}		0.011^{***}	-0.485^{***}		-0.119^{***}
	(0.001)		(0.001)	(0.011)		(0.010)
Duration (T)		0.437^{***}	0.438^{***}		0.464^{***}	0.457^{***}
		(0.001)	(0.001)		(0.003)	(0.003)
Daily Volatility (σ_D)	0.930^{***}	0.911^{***}	0.910^{***}	0.908^{***}	0.874^{***}	0.874^{***}
	(0.003)	(0.003)	(0.003)	(0.007)	(0.007)	(0.007)
Observations	410,792	410,792	410,792	102,698	102,698	102,698
R^2	0.148	0.343	0.343	0.139	0.335	0.336
Adjusted R^2	0.148	0.343	0.343	0.139	0.335	0.336
Residual Std. Error	1.437	1.262	1.262	1.481	1.301	1.300
F Statistic	$35{,}724.510^{***}$	$107{,}191.000^{***}$	$71{,}508.110^{***}$	$8{,}316^{***}$	$25{,}892.110^{***}$	$17{,}328.950^{***}$
AIC	1,463,712	1,357,075	1,356,983	372,049.6	345,533.1	345,400.1

Note: *p<0.1; **p<0.05; ***p<0.01

In the second regression for all trades (column 3), the estimated regression coefficient on trade duration, $\hat{\beta}_d$, is 0.437, and is statistically significant at the 1% level, has already found in section 4.4. When both the participation rate and trade duration are included in the regression (column 4), the coefficient on trade duration remains close to 1/2 (specifically 0.438) while the regression coefficient on the participation rate becomes positive 0.011. Both regression coefficients are statistically significant at the 1% level However, despite being statistically significant, the estimated regression coefficient on the daily volume fraction is very small and it implies that a 1% increase in daily volume fraction leads to only a 0.011% increase in the price change. Moreover, the coefficient of determination compared is the same (34.3%) as in the second regression.

The findings are even more striking if we focus only on large trades. The estimated regression coefficient $\hat{\beta}_p$ from the first regression is -0.485 (column 5), and is statistically significant at the 1% level, confirming a negative relationship between the participation rate and price changes. When both the average participation rate and trade duration are included in the regression (column 7), the coefficient on trade duration remains close to 1/2 (specifically 0.457) and the regression coefficient on the participation rate remains negative at -0.119. Both regression coefficients are statistically significant at the 1% level, and the inclusion of the daily volume fraction as a statistically significant explanatory variable is confirmed when we compare the nested models in equations (37) and (38), with an ANOVA test (F- statistic of 191.4, p-value: $< 2.2\ 10^{-16}$) or using the AIC. However the estimated regression coefficient on the participation rate is very small: a 1% increase in daily volume fraction leads to only a 0.071% increase in the price change, showing that trade duration remains the main determinant of the amplitude of price variation.

These results confirm that, in orderly executions, an increase in the average participation rate at constant trade duration leads to a *decrease* in price changes. Or, in other words, an *increase* in the trading speed leads to a *decrease* in the amplitude of price changes during execution, similar to the findings of Salem et al. (2018) in the US Treasury market. This is simply because a faster execution entails a shorter duration therefore a lower exposure to volatility risk during execution. Note that this is the opposite effect to what is usually assumed in many optimal execution models.

6. Conclusion

A generation of models has emphasised the importance of market 'impact' as the main determinant of execution costs and defined optimal execution in terms of a tradeoff between 'market impact' and price volatility. Our empirical analysis suggests that in fact 'market impact' and price volatility are two sides of the same coin. We have shown that price changes are mainly driven by the *aggregate (net) order flow imbalance* rather than the 'impact' of trades and that, once conditioned on trade duration, trade size has very little explanatory power for price changes during execution. We have found that price changes are explained by the square-root of *duration* and that this relationship arises from the well-known square-root scaling of volatility as a function of time. Further, conditional on trade size, the amplitude of price changes is shown to increases with duration, contradicting previous claims regarding the 'invariance' of impact with respect to trade duration. When looking at the role of the participation rate in determining market 'impact', we have found that for trades with high participation rates, mainly executed on a VWAP schedule, a large increase in the trade size (and in the participation rate) leaves prices almost unchanged, contradicting one of the key assumptions of optimal execution models. Finally, our analysis of the relationship between price changes and the average participation rate *conditional* on trade duration shows that the trading speed affects price changes in the opposite direction than what has been assumed in optimal execution models: the *slower* the execution the *higher* the price variation during the trade.

Our findings bring into question the notion of 'market impact' defined at the trade level and highlight the need to rethink what it means to optimally execute a trade. Our results suggest that it is more meaningful to focus on the modelling of aggregate order flow dynamics and the management of portfolio volatility during execution rather than the optimisation of 'impact' at a trade-by-trade level.

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